Does DMSP Nighttime Light Capture Local Economic Welfare? Evidence from a Cluster-Level Quasi-Panel of Sub-Saharan Africa

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Abstract

Recently, an increasing number of researchers have used DMSP and VIIRS nightlight data to measure economic activity in developing countries, where GDP and income data are scarce. However, several researchers have begun to question the quality of DMSP nightlight data at the subnational level and at rural area.

By constructing a cluster-level quasi-panel using Demographic and Health Survey data, we examine how changes in nightlight intensity correlate with household welfare. Our findings indicate that cluster-level nightlight is strongly correlated with household living standards, such as fuel type, floor type, and ownership of motorcycles and bicycles, once the noise in nightlight data is accounted for. This suggests that nightlight data can be a useful proxy for measuring household-level welfare.

1 Introduction

Recently, an increasing number of researchers have used nightlight data to measure economic activity in developing countries, where GDP and income data are scarce (Joseph, 2022; Bruederle and Hodler, 2018; Jhamb et al., 2025; Weidmann and Schutte, 2017; Fontaine et al., 2025).[1] However, several researchers have begun to question the quality of DMSP nightlight data at the subnational level, in rural areas, and at the micro level (Gibson et al., 2021; Keola et al., 2015; Chen and Nordhaus, 2011; Bluhm and Krause, 2022), cautioning against the use of nighttime lights as a proxy for economic activity at local scales.

Gibson et al. (2021) compare DMSP and VIIRS nightlights data to assess whether such data should be used in economic research at the rural and urban level, and find that both DMSP and VIIRS nightlights are poor proxies for regional GDP; they recommend conducting separate analyses for urban and rural areas. According to Keola et al. (2015), the premise that economic activity emits more nighttime light does not hold uniformly across sectors: luminosity may rise sharply in industrial or service clusters where economic activity is spatially concentrated, but agriculture and forestry, which dominate output in much of rural Sub-Saharan Africa, may not generate sufficient nighttime lighting to reflect underlying economic growth.

We argue that this conclusion is not as general as it appears, and that it reflects specific empirical and conceptual choices rather than a fundamental limitation of nighttime-lights data. We identify four challenges in evaluating the localized relationship between nightlights and economic development. First, Gibson et al. (2021) evaluate nighttime lights against Indonesian regency-level GDP, a context that differs fundamentally from Sub-Saharan Africa in ways that limit the generalizability of their findings. Indonesia is a middle-income country with a well-established national statistical system and relatively standardized subnational GDP accounting, a higher baseline level of rural electrification, and a different sectoral and spatial structure of economic activity than the predominantly rural, low-luminosity, and less electrified economies of Sub-Saharan Africa. A finding that DMSP lights perform poorly as

[1]There are several sources of nightlights such as the United States Air Force Defense Meteorological Satellite Program (DMSP), the Visible Infrared Imaging Radiometer Suite (VIIRS), etc.

a proxy for second-level administrative GDP in Indonesia does not straightforwardly transfer to a setting where baseline luminosity is much lower, rural non-farm economic activity is structured differently, and the relevant outcome is not aggregate regional GDP but household-level welfare measured at the village cluster scale. Second, regional GDP may not be an appropriate benchmark for living standards or economic welfare in the first place, since non-monetary consumption, subsistence production, and asset accumulation constitute important components of welfare, particularly in rural economies, and are by construction poorly captured by GDP. Third, regional GDP is an aggregate administrative-unit measure that may not adequately capture household-level welfare outcomes, the level at which nightlights are most often applied in practice; our analysis instead links nighttime lights measured within a 15-kilometer radius of each DHS cluster directly to household asset ownership, housing quality, and service access, a considerably finer and more welfare-relevant geographic scale than the regency-level GDP used by Gibson et al. (2021). Fourth, their empirical design relies on only two years of DMSP data (2011 and 2012) and two years of VIIRS data (2015 and 2016), a very short window evaluated with only time fixed effects and without area-level fixed effects or controls for local population; such a specification cannot account for unobserved local characteristics or exploit within-location changes in luminosity over time, and is especially vulnerable to cross-sectional confounding in rural, low-luminosity settings where the lights signal is faint to begin with. As a result, the inability of nighttime lights to explain cross-sectional variation in regional GDP at the regency level in Indonesia does not necessarily imply that nighttime lights fail to capture changes in household economic welfare over time at the village cluster level in Sub-Saharan Africa; it may instead reflect a mismatch between the geographic and conceptual scale at which lights are evaluated and the scale at which their signal is most meaningful, compounded by a specification that lacks the within-location identification needed to isolate that signal. We also note that the specific data artifacts most often cited against DMSP — blooming, top-coding, and the absence of on-board radiometric calibration — are concerns that originate primarily in the remote-sensing literature on high-luminosity urban contexts; in Sub-Saharan Africa, where the overwhelming majority of DHS clusters are rural and low-luminosity to begin with, these particular distortions are likely to be considerably less consequential than in the contexts where they were first identified.

To address these challenges, we use household living standards rather than regional GDP as our outcome variables. We draw on georeferenced DHS microdata containing detailed information on household assets, including ownership of radios, televisions, bicycles, and motorcycles, as well as housing quality, sanitation, electricity access, and cooking fuel use, all of which more directly reflect local economic welfare and living standards than an aggregate GDP figure. We construct a cluster-level quasi-panel by linking DHS clusters to DMSP nighttime lights within a 15-kilometer radius of each cluster for the period 2000–2020, enabling the analysis to exploit both spatial and temporal variation in luminosity at fine geographic scales.

We construct this quasi-panel using the GPS coordinates of each cluster (village). We designate the earliest available survey as the initial wave, and match each cluster in subsequent waves to a cluster in the initial wave if it is located within 10 kilometers. To do so, we first calculate the distance between clusters in each pair of initial and subsequent waves; second, we select the closest cluster within the 10 km radius as the matched cluster when multiple candidates exist; third, we verify that each matched cluster is also the closest to its pair, in order to construct a consistent cluster-level panel. Clusters in the initial wave without a corresponding match in later waves are excluded from the analysis. Our sample is limited to 17 Sub-Saharan African countries with at least two DHS survey waves that include GPS data, yielding a quasi-panel of 5,217 clusters, with cluster-level nightlights measured within a 15 km radius using a five-year average centered on the interview year of each DHS wave.

Our findings indicate that nighttime lights are strongly associated with multiple dimensions of household welfare. Using a cluster-level quasi-panel with cluster and period fixed effects, we find that increases in nighttime light intensity are positively and significantly associated with household access to electricity, ownership of radios, televisions, and bicycles, as well as improved sanitation and the use of clean cooking fuels. These relationships remain robust after controlling for cluster population and household characteristics, and are largely unchanged when country-by-period fixed effects are included. The one exception is motorcycle ownership, for which we find no statistically significant relationship. Critically, when we re-estimate the model separately for rural and urban subsamples, the relationship between nighttime lights and household welfare remains positive and statistically significant in *both* samples, directly contradicting the central claim in Gibson et al. (2021) that night-lights are a poor proxy specifically in rural areas. This result suggests that the weak rural performance reported in prior cross-sectional work reflects the limitations of that empirical design rather than a genuine absence of signal in rural luminosity data. Overall, our results indicate that DMSP nighttime lights capture meaningful variation in local household welfare and can serve as a useful proxy for economic development at the local level in Sub-Saharan Africa, in both rural and urban settings alike.

A related literature has examined whether nighttime lights can serve as a proxy for local economic development and welfare more broadly. Using georeferenced DHS data from 29 African countries, Bruederle and Hodler (2018) show that DMSP-OLS nighttime lights are strongly associated with local measures of wealth, education, and health at both the DHS cluster and grid-cell level, suggesting that nighttime lights can capture human development at highly disaggregated spatial scales, though their specification controls only for country-by-time fixed effects, without cluster or time fixed effects, and does not separately examine rural and urban areas. Similarly, Weidmann and Schutte (2017) combine DMSP-OLS night-time lights with DHS surveys from 37 African countries and find a strong positive relationship between nighttime lights and household wealth, though their contribution is primarily a prediction and visualization exercise rather than a regression framework designed to isolate the lights-welfare relationship net of confounding factors. More recently, Jhamb et al. (2025) employ a harmonized DMSP-VIIRS nighttime lights dataset together with cluster-level DHS data to show that nighttime lights are strongly correlated with GDP per capita, wealth, and human development indicators, reporting a stronger relationship in urban than in rural areas; however, their identification still rests substantially on cross-country variation, and their specification does not include cluster fixed effects. Despite these contributions, existing studies primarily focus on wealth indices, GDP, or broad human development indicators. In contrast, this study examines whether DMSP nighttime lights capture multiple dimensions of household welfare, including asset ownership, housing quality, electricity access, sanitation, and cooking fuel use, using a cluster-level quasi-panel constructed from DHS surveys across 17 Sub-Saharan African countries, and explicitly tests, rather than assumes, whether the relationship holds separately in rural and urban areas.

This study contributes to the growing literature on nighttime lights and economic development in four main ways. First, unlike most existing studies, which rely on cross-sectional variation or only one to two years of data, we construct a cluster-level quasi-panel from geo-referenced DHS surveys across 17 Sub-Saharan African countries conducted during 2000–2020, allowing us to exploit both spatial and temporal variation in nighttime lights over a much longer horizon. Second, rather than focusing on GDP, wealth indices, or broad development indicators, we examine multiple dimensions of household welfare, including asset ownership, housing quality, electricity access, sanitation, and cooking fuel use, measures that may more accurately represent living standards, particularly in rural areas where subsistence production and non-monetary consumption are important components of economic welfare. Third, we employ a panel fixed-effects framework controlling for cluster fixed effects, period fixed effects, and country-by-period fixed effects, together with local population, an empirical specification substantially richer than that used in the studies most closely related to ours, none of which jointly control for all of these factors. Fourth, by explicitly estimating the lights-welfare relationship separately for rural and urban subsamples and finding it positive and significant in both, we directly test, and reject, the claim that nighttime lights are an unreliable proxy in rural areas, providing evidence that qualifies rather than confirms the more pessimistic conclusions of recent critiques of the DMSP data.

This paper is organized as follows. Section 2 elaborates the estimation strategy. Section 3 describes the construction of the dataset. Section 4 presents the results, and Section 5 offers concluding remarks.

2 Data Constraints and the Use of Nighttime Lights

The limitations of measuring economic growth have long been a central topic of debate in economics. First, there are widespread concerns about the accuracy of traditional data sources. Survey- and census-based data provided by national statistical offices are often imperfect, and in some cases may be intentionally manipulated to misrepresent a country's economic conditions. Other factors, such as conflict and corruption, can also contribute to poor data quality. Second, reliable economic data are often unavailable for many low- and middle-income countries. Third, most measures of economic growth are reported only at the national level, with limited or no information available at finer geographic scales. Fourth, even when census-based data are available, they are typically collected only once every five or ten years. As a result, the lack of annually available data poses an additional challenge for measuring and tracking economic growth over time.

Since 1992, with the emergence of satellite-based nighttime lights data, many researchers have adopted nighttime lights data as an alternative measure of economic activity and economic growth (Henderson et al., 2012; Chen and Nordhaus, 2011; Storeygard, 2016; Elvidge et al., 1997; Yousafzai and Naito, 2025). First, satellite data can be applied to specific geographic areas at various spatial scales. Second, they capture variation over time at both national and subnational levels. These characteristics make satellite remote-sensing data an accessible and cost-effective source for economic analysis with high temporal and spatial resolution. Satellite data on nighttime luminosity reflect socioeconomic development in human settlements and serve as a reliable alternative proxy for economic activity. Building on these advantages, a growing body of research has explored the use of satellite-based measures of economic activity. Pérez-Sindín et al. (2021) demonstrate that light data are strong indicators for estimating changes in socioeconomic development at the subnational level in middle- and low-income countries. Similarly, Chen and Nordhaus (2011) show that nighttime lights provide valuable statistical information for countries with low-quality statistical systems.

Most of studies that suggest to use nightlights as a proxy for income discussed the accuracy limitations of the DMSP nightlights data including blurred features, coarse spatial resolution, reflections from snow and water that are falsely captured as lights, top-coding that prevents measuring growth over time, and a lack of onboard calibration. They also reveal that VIIRS nightlight data are better in capturing more localized effect.

3 Estimation Strategy

To quantify the effect of nighttime light intensity on household welfare, we estimate the following specification:

$$Y_{hcjt} = \alpha_0 + \alpha_1 \ln(NL_{cjt}) + \alpha_2 \ln(Population_{cjt}) + \alpha_3 Z_{hcjt} + \eta_c + \eta_t + \varepsilon_{hcjt}, \tag{1}$$

where h, c, j, and t denote the household, cluster, country, and period indices, respectively, with period measured in five-year intervals. We compile DHS data from 17 countries and construct a cluster (village)-level quasi-panel based on GPS coordinates, assigning each cluster a unique index c. After constructing the quasi-panel, we draw a 15-kilometer buffer around each cluster's base-wave location and calculate the average nighttime light intensity within that buffer for each five-year period.

The outcome variable, Y_{hcjt}, is a binary indicator for household h equal to one if the household owns a given asset, has adequate housing conditions, or has improved living conditions (defined precisely in Section 4.2), and zero otherwise; because the outcome is binary, we estimate a linear probability model rather than log-transforming Y_{hcjt}. The key explanatory variable, $\ln(NL_{cjt})$, is the natural logarithm of the five-year average of nighttime light intensity in cluster c. We additionally control for $\ln(Population_{cjt})$, the natural logarithm of cluster-level population. The vector Z_{hcjt} comprises household-level control variables: the household head's completed years of schooling, age, and gender, as well as household size.

We include cluster fixed effects, η_c, which absorb unobserved, time-invariant characteristics specific to each cluster, and period fixed effects, η_t, which capture period-specific shocks common to all clusters. The error term is denoted ε_{hcjt}. Standard errors are clustered at the subnational level (DHS geolevel2 administrative units) to allow for correlation of error terms within each subnational unit, both across households and over time.

As a robustness check, we further extend the baseline specification by including country $\times$ period fixed effects, η_{jt}, to account for time-varying unobserved heterogeneity at the country level, such as macroeconomic shocks or policy changes that may simultaneously affect nighttime light intensity and household welfare outcomes.

4 Dataset Construction

4.1 Construction of a Cluster-Level Quasi-Panel

The Demographic and Health Survey (DHS) is used to measure household living standards. Since the DHS is not a panel dataset, we construct a quasi-panel using the GPS coordinates of each cluster (village). Specifically, we designate the earliest available survey wave as the initial wave. Each cluster in a subsequent wave is matched to a cluster in the initial wave if it is located within 10 kilometers. To do so, we proceed in three steps. First, we calculate the distance between every cluster in the initial wave and every cluster in each subsequent wave. Second, when multiple candidate clusters lie within the 10-kilometer radius, we select the nearest one as the matched cluster. Third, we verify that the match is mutual—that is, the matched cluster in the initial wave is also the nearest neighbor to the corresponding cluster in the subsequent wave—to ensure a consistent cluster-level panel. Clusters in the initial wave without a corresponding match in later waves are excluded from the analysis. Our sample is limited to Sub-Saharan African countries with at least two DHS survey waves that include GPS data.[2]

4.2 Construction of Welfare and Nighttime Lights Indicators

We use electricity access, cooking fuel type, floor material, sanitation, and ownership of motorcycles, bicycles, radios, and televisions as proxies for household welfare. Electricity access and ownership of radios, televisions, bicycles, and motorcycles are treated as binary outcome variables, equal to 1 if a household has access to or owns the item, and 0 otherwise. The variables for cooking fuel, floor material, and sanitation are also binary, constructed from categorical survey responses. The improved cooking fuel variable equals 1 if a household uses electricity, LPG, natural gas, or biogas, and 0 otherwise. The improved floor material variable equals 1 if the household floor is made of wood planks, polished wood, ceramic

[2]Table A1 in the Supplementary Information presents the list of Sub-Saharan African countries and DHS waves included in the analysis.

tiles, carpet, or cement, and 0 otherwise. The improved sanitation variable equals 1 if a household uses a flush toilet, and 0 otherwise.[3]

We use nighttime lights data from the United States Air Force Defense Meteorological Satellite Program (DMSP), at a spatial resolution of 1 km $\times$ 1 km. The original DMSP nighttime lights series covers the period 1992–2012; NOAA has since extended this series to cover 2013–2021. The data are available as pixels of 30 $\times$ 30 arc seconds, corresponding to less than one square kilometer at the equator. Each pixel reports an annual light intensity value as a digital number (DN) ranging from 0 to 63, where a DN of 0 indicates no detectable light and a DN of 63 indicates the maximum recorded light intensity. In developed countries or brightly lit urban areas, entire cities may register a DN of 63, reflecting a top-coding (saturation) problem in the DMSP sensor.

To construct the aggregated nighttime lights measure at the cluster level, we proceed in three steps. First, we link the geo-referenced DHS cluster locations to the nighttime lights satellite data. Second, we define the spatial unit of analysis as a 15-kilometer-radius buffer around each cluster, and calculate nighttime light intensity within this buffer by summing the digital numbers of all pixels it contains, for each satellite-year over the period 1997–2021. Across our full analysis period, we have a total of 41 satellite-year pairs. For most years, nighttime lights data are recorded by two satellites; for years covered by both satellites, we take the average of the two values. Third, we average nighttime light intensity over the five-year window corresponding to each DHS interview year. For example, for a DHS survey wave with an interview year of 2010, we sum the digital numbers within the 15-kilometer buffer for each year from 2008 to 2012, average across these five years, and assign the resulting value to interview year 2010.

This procedure yields a total of 5,217 georeferenced DHS clusters (villages) from survey waves conducted between 2000 and 2020 across the selected countries. To construct a cluster-level quasi-panel, we retain only countries with at least two DHS survey waves containing GPS coordinates during the study period. Consequently, we exclude Togo, Côte d'Ivoire, and Gabon because they do not satisfy this requirement, and we also exclude Liberia

[3] We use DHS survey waves conducted in or after 2000 because most surveys conducted before 2000 do not include information on the improved cooking fuel variable.

due to the disruption caused by its prolonged civil war.

We also construct a measure of cluster-level population, following the same procedure used to construct the nighttime lights variable. We extract population counts within a 15-kilometer radius around each cluster from the Gridded Population of the World (GPWv4), obtained from the Socioeconomic Data and Applications Center (SEDAC) Center for International Earth Science Information Network–CIESIN–Columbia University (2018). This spatially disaggregated dataset is available in five-year intervals (2000, 2005, 2010, 2015, and 2020) at a 30-arc-second resolution (approximately one kilometer at the equator).

To identify subnational administrative units consistently across countries and survey waves, we use the harmonized second-level administrative boundary dataset provided by IPUMS International.[4] This dataset contains a globally harmonized shapefile of second-level administrative units and corresponding geographic identifiers (GEOLEV2). Using the geographic coordinates of DHS clusters, we spatially match each cluster to its corresponding GEOLEV2 administrative unit. The resulting geolevel2 identifier is then merged with the cluster-level quasi-panel dataset and used to cluster standard errors, allowing for arbitrary correlation of error terms within subnational administrative units over time.

5 Empirical Results

5.1 Descriptive Statistics

Table 1 presents descriptive statistics for the full sample of 326,741 household observations. The data combine georeferenced household survey information with satellite-derived nighttime lights measured at the cluster level using a 15 km radius buffer around each cluster.

The mean value of nighttime lights is 4,984, with substantial variation across locations (standard deviation = 5,592), indicating significant spatial heterogeneity in economic activity. When expressed in logarithmic form, nighttime lights have a mean of 8.16, reflecting the skewed distribution of luminosity across regions. Nighttime lights per capita also exhibit considerable dispersion, with an average of 0.044 and a maximum value exceeding 2.2,

[4]The GEOLEV2 subnational administrative boundary dataset can be obtained from the following link: https://international.ipums.org/international/gis_harmonized_2nd.shtml

Table 1. Summary Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Nightlights and Geography					
Nighttime Lights	326741	4984.458	5592.491	798.201	38619.094
Ln Nighttime Lights	326741	8.161	0.735	6.682	10.562
Cluster Population	326741	433847.16	741363	884.849	4474908.5
Nighttime Lights per Capita	326741	0.044	0.086	0.002	2.238
Ln Nighttime Lights per Capita	326741	-3.811	1.066	-6.104	0.805
Household welfare outcomes					
Electricity Access	326741	0.325	0.468	0	1
Radio Ownership	326741	0.593	0.491	0	1
Television Ownership	326741	0.261	0.439	0	1
Motorcycle Ownership	326741	0.099	0.299	0	1
Bicycle Ownership	326741	0.25	0.433	0	1
Improved Floor Materials	326741	0.088	0.283	0	1
Improved Sanitation Access	326741	0.116	0.32	0	1
Clean Cooking Fuel Use	326741	0.098	0.297	0	1
Household characteristics					
Household Head Age	326741	44.664	15.773	9	94
Household Head Gender	326741	0.71	0.454	0	1
Household Head Schooling (Years)	326741	4.989	4.938	0	25
Household Size	326741	4.694	2.444	1	11
Urban	326741	0.395	0.489	1	11

Notes: Urban is equal to one if the cluster is located in an urban area in the initial wave of the DHS survey. This data contains 5,217 clusters (villages) in the analysis.

suggesting pronounced differences in local economic density.

Turning to household welfare outcomes, 32.5 percent of households report having access to electricity. Ownership rates for consumer durables vary widely, with 59.3 percent of households owning a radio and 26.1 percent owning a television. Ownership of transport assets remains relatively low, with 9.9 percent of households owning a motorcycle and 25 percent owning a bicycle.

In terms of housing quality and basic services, only 8.8 percent of households live in dwellings with improved floor materials, while 11.6 percent have access to improved sanitation. Clean cooking fuel use remains limited at 9.8 percent, highlighting persistent energy poverty in the sample.

Household heads are, on average, 44.7 years old, and 71 percent are male. The average years of schooling among household heads is approximately five years, while the average household size is 4.7 members.

Overall, the descriptive patterns indicate substantial heterogeneity in both nighttime light intensity and household welfare indicators, which motivates the empirical analysis of whether spatial variation in economic activity is systematically associated with differences in household living standards.

5.2 Main Results

This section presents the main empirical results based on the specification outlined in Equation (1), which estimates the relationship between nighttime lights intensity and household welfare outcomes.

Table 2 presents the estimated effects of nighttime lights on household access to electricity. Column (1) presents a simple OLS estimate without controls. Column (2) introduces cluster fixed effects, while Column (3) adds period fixed effects. Columns (4)–(6) progressively introduce cluster-level population, household head characteristics, and household size, respectively. The final specification in Column (6) includes all controls and fixed effects and is considered the preferred specification.

Table 2 shows that nighttime lights are positively associated with household access to electricity. In Column 6, the estimated coefficient is positive and statistically significant at the 1 percent level. It implies that a one percent increase in the nighttime lights is associated with an approximately 8.2 percentage point increase in the probability of having electricity access.

Table 3 shows that nighttime lights are positively and significantly associated with household radio ownership across all specifications. In Column 6, the estimated coefficient shows that a one percent increase in the nightlight intensity is associated with a 9.8 percentage point increase in the probability of radio ownership. The coefficient remains stable after adding fixed effects and controls, indicating a robust relationship. The increase in explanatory power

Table 2. The Effects of Nighttime Lights on Household Access to Electricity

	Household Electricity Access					
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Ln Lights	0.280***	-0.0825***	0.0939***	0.0831***	0.0818***	0.0824***
	(0.0142)	(0.0199)	(0.0263)	(0.0241)	(0.0219)	(0.0219)
Ln Population				✓	✓	✓
Household Head Controls					✓	✓
Household Size						✓
Number of Observations	326,741	326,741	326,741	326,741	326,741	326,741
R-squared	0.193	0.517	0.543	0.543	0.572	0.573
Number of Clusters	5217	5217	5217	5217	5217	5217
Number of Countries	17	17	17	17	17	17
Cluster Fixed Effects	–	Yes	Yes	Yes	Yes	Yes
Period Fixed Effects	–	–	Yes	Yes	Yes	Yes

Note: Standard errors are clustered at the subnational (geolevel2) level and reported in parentheses.
*** p<0.01, ** p<0.05, * p<0.1.

Table 3. The Effects of Nighttime Lights on Household Radio Ownership

	Household Radio Ownership					
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Ln Lights	0.111***	0.156***	0.0948***	0.0999***	0.0953***	0.0980***
	(0.00770)	(0.0115)	(0.0166)	(0.0155)	(0.0166)	(0.0163)
Ln Population				✓	✓	✓
Household Head Controls					✓	✓
Household Size						✓
Observations	326,741	326,741	326,741	326,741	326,741	326,741
R-squared	0.028	0.127	0.132	0.132	0.177	0.184
Number of Clusters	5217	5217	5217	5217	5217	5217
Number of Countries	17	17	17	17	17	17
Cluster Fixed Effects	–	Yes	Yes	Yes	Yes	Yes
Period Fixed Effects	–	–	Yes	Yes	Yes	Yes

from an R-squared of 0.028 in the baseline model to 0.184 in the full specification further indicates that fixed effects and controls substantially improve model fit.

Table 4 implies that nighttime lights are positively and significantly associated with household television ownership. In Column 6, the estimated coefficient shows that a one

Table 4. The Effects of Nighttime Lights on Household Television Ownership

	Household Television Ownership					
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Ln Lights	0.239***	-0.00426	0.118***	0.110***	0.108***	0.111***
	(0.00908)	(0.0171)	(0.0236)	(0.0223)	(0.0187)	(0.0193)
Ln Population				✓	✓	✓
Household Head Controls					✓	✓
Household Size						✓
Observations	326,741	326,741	326,741	326,741	326,741	326,741
R-squared	0.160	0.378	0.401	0.401	0.446	0.453
Number of Clusters	5217	5217	5217	5217	5217	5217
Number of Countries	17	17	17	17	17	17
Cluster Fixed Effects	–	Yes	Yes	Yes	Yes	Yes
Period Fixed Effects	–	–	Yes	Yes	Yes	Yes

Table 5. The Effects of Nighttime Lights on Household Bicycle Ownership

	Household Bicycle Ownership					
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Ln Lights	-0.0375***	0.0783***	0.0426***	0.0519***	0.0485***	0.0518***
	(0.0113)	(0.00862)	(0.0151)	(0.0131)	(0.0133)	(0.0132)
Ln Population				✓	✓	✓
Household Head Controls					✓	✓
Household Size						✓
Observations	326,741	326,741	326,741	326,741	326,741	326,741
R-squared	0.004	0.272	0.273	0.273	0.296	0.309
Number of Clusters	5217	5217	5217	5217	5217	5217
Number of Countries	17	17	17	17	17	17
Cluster Fixed Effects	–	Yes	Yes	Yes	Yes	Yes
Period Fixed Effects	–	–	Yes	Yes	Yes	Yes

Note: Standard errors are clustered at the subnational (geolevel2) level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

percent increase in nightlight intensity is associated with an 11.1 percentage point increase in the probability of television ownership.

Table 5 indicates that nighttime lights are positively and significantly associated with household bicycle ownership. In Column 6, a one-unit increase in the logarithm of night-

Table 6. The Effects of Nighttime Lights on Household Motorcycle Ownership

	Household Motorcycle Ownership					
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Ln Lights	0.0234	0.0125	0.0200	0.0110	0.00953	0.0108
	(0.0186)	(0.00812)	(0.0176)	(0.0161)	(0.0168)	(0.0166)
Ln Population				✓	✓	✓
Household Head Controls					✓	✓
Household Size						✓
Observations	326,741	326,741	326,741	326,741	326,741	326,741
R-squared	0.003	0.282	0.288	0.289	0.303	0.307
Number of Clusters	5217	5217	5217	5217	5217	5217
Number of Countries	17	17	17	17	17	17
Cluster Fixed Effects	–	Yes	Yes	Yes	Yes	Yes
Period Fixed Effects	–	–	Yes	Yes	Yes	Yes

Note: Standard errors are clustered at the subnational (geolevel2) level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 7. The Effects of Nighttime Lights on Improved Floor Materials

	Improved Floor Dummy					
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Ln Lights	0.0877***	-0.0773***	0.0409	0.0339	0.0334	0.0337
	(0.0112)	(0.0163)	(0.0327)	(0.0287)	(0.0268)	(0.0268)
Ln Population				✓	✓	✓
Household Head Controls					✓	✓
Household Size						✓
Observations	326,741	326,741	326,741	326,741	326,741	326,741
R-squared	0.052	0.246	0.260	0.260	0.283	0.284
Number of Clusters	5217	5217	5217	5217	5217	5217
Number of Countries	17	17	17	17	17	17
Cluster Fixed Effects	–	Yes	Yes	Yes	Yes	Yes
Period Fixed Effects	–	–	Yes	Yes	Yes	Yes

Note: Standard errors are clustered at the subnational (geolevel2) level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

time lights is associated with a 5.18 percentage point increase in the probability of bicycle ownership.

Table 6 indicates that nighttime lights are positively associated with household motorcycle ownership. In Column 6, a one percent increase in the nighttime lights intensity is associated with a 1.08 percentage point increase in the probability of motorcycle ownership, but the estimate is not statistically different from zero. These results suggest that nighttime lights do not have a robust relationship with household motorcycle ownership.

Table 8. The Effects of Nighttime Lights on Clean Cooking Fuel Use

	Clean Cooking Fuel Use					
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Ln Lights	0.133***	-0.00193	0.0794***	0.0781***	0.0775***	0.0770***
	(0.0222)	(0.00977)	(0.0302)	(0.0295)	(0.0276)	(0.0275)
Ln Population				✓	✓	✓
Household Head Controls					✓	✓
Household Size						✓
Observations	326,741	326,741	326,741	326,741	326,741	326,741
R-squared	0.108	0.424	0.431	0.431	0.460	0.461
Number of Clusters	5217	5217	5217	5217	5217	5217
Number of Countries	17	17	17	17	17	17
Cluster Fixed Effects	–	Yes	Yes	Yes	Yes	Yes
Period Fixed Effects	–	–	Yes	Yes	Yes	Yes

Note: Standard errors are clustered at the subnational (geolevel2) level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 7 shows that nighttime lights are positively associated with improved floor materials. In Column 6, the estimated coefficient is positive but statistically insignificant. It implies that a one percent increase in the nighttime lights is associated with an approximately 3.4 percentage point increase in the probability of having improved floor materials.

Table 8 shows that nighttime lights are positively associated with clean cooking fuel use. In Column 6, the estimated coefficient is positive and statistically significant at the 1 percent level. It implies that a one percent increase in the nighttime lights is associated with an approximately 7.7 percentage point increase in the probability of clean cooking fuel use.

Table 9 shows that nighttime lights are positively associated with improved sanitation access. In Column 6, the estimated coefficient is positive and statistically significant at the 1 percent level. It implies that a one percent increase in nighttime lights is associated with an

Table 9. The Effects of Nighttime Lights on Improved Sanitation Access

	Improved Sanitation Access					
Variable	(1)	(2)	(3)	(4)	(5)	(6)
Ln Lights	0.153***	-0.0147*	0.0919***	0.0911***	0.0906***	0.0906***
	(0.0244)	(0.00870)	(0.0161)	(0.0152)	(0.0140)	(0.0141)
Ln Population				✓	✓	✓
Household Head Controls					✓	✓
Household Size						✓
Observations	326,741	326,741	326,741	326,741	326,741	326,741
R-squared	0.124	0.441	0.449	0.449	0.470	0.470
Number of Clusters	5217	5217	5217	5217	5217	5217
Number of Countries	17	17	17	17	17	17
Cluster Fixed Effects	–	Yes	Yes	Yes	Yes	Yes
Period Fixed Effects	–	–	Yes	Yes	Yes	Yes

Note: Standard errors are clustered at the subnational (geolevel2) level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 10. The Effects of Nighttime Lights on Household Access to Electricity: Urban Sample

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Outcome Variables	Household Electricity Access	Household Radio Ownership	Household Television Ownership	Household Bicycle Ownership	Household Motorcycle Ownership	Improved Floor Materials	Clean Cooking Fuel Use	Improved Sanitation Access
Ln Lights	0.0491	0.103***	0.0697***	0.0716***	0.0281	-0.0284	0.113***	0.0937***
	(0.0364)	(0.0227)	(0.0233)	(0.0140)	(0.0278)	(0.0315)	(0.0417)	(0.0250)
Ln Population	✓	✓	✓	✓	✓	✓	✓	✓
Household Head Controls	✓	✓	✓	✓	✓	✓	✓	✓
Household Size	✓	✓	✓	✓	✓	✓	✓	✓
Number of Observations	129,033	129,033	129,033	129,033	129,033	129,033	129,033	129,033
R-squared	0.442	0.142	0.351	0.244	0.329	0.294	0.463	0.454
Number of Clusters	2,548	2,548	2,548	2,548	2,548	2,548	2,548	2,548
Number of Countries	17	17	17	17	17	17	17	17
Cluster Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: Standard errors are clustered at the subnational (geolevel2) level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

approximately 9.1 percentage point increase in the probability of having access to improved sanitation facilities.

Table 10 presents the relationship between nighttime light intensity and household welfare indicators for the urban sample. The regressions include cluster fixed effects and survey period fixed effects while controlling for population size, household head characteristics, and household size.

Table 11. The Effects of Nighttime Lights on Household Access to Electricity: Rural Sample

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Outcome Variables	Household Electricity Access	Household Radio Ownership	Household Television Ownership	Household Bicycle Ownership	Household Motorcycle Ownership	Improved Floor Materials	Clean Cooking Fuel Use	Improved Sanitation Access
Ln Lights	0.0802***	0.123***	0.0517***	0.0595***	0.0123	0.0134**	0.0100	0.0467***
	(0.0151)	(0.0138)	(0.00946)	(0.0106)	(0.00857)	(0.00562)	(0.00940)	(0.0130)
Ln Population	✓	✓	✓	✓	✓	✓	✓	✓
Household Head Controls	✓	✓	✓	✓	✓	✓	✓	✓
Household Size	✓	✓	✓	✓	✓	✓	✓	✓
Number of Observations	197,708	197,708	197,708	197,708	197,708	197,708	197,708	197,708
R-squared	0.384	0.183	0.272	0.350	0.288	0.201	0.240	0.271
Number of Clusters	3,770	3,770	3,770	3,770	3,770	3,770	3,770	3,770
Number of Countries	17	17	17	17	17	17	17	17
Cluster Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: Standard errors are clustered at the subnational (geolevel2) level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

The results indicate that increases in nighttime light intensity are positively associated with several dimensions of household welfare in urban areas, with the exception of improved floor materials. Most of the estimated coefficients are also statistically significant. However, the estimated coefficients for household electricity access, motorcycle ownership, and improved floor materials are statistically insignificant. In addition, the coefficient on improved floor materials is negative, although it is not statistically significant. One possible explanation is that these indicators exhibit relatively limited variation within urban areas. Electricity coverage is already relatively high in many urban locations, leaving little scope for further improvements as local economic activity increases. Likewise, housing quality and motorcycle ownership may depend more on long-term household wealth, housing market conditions, or local institutional factors than on short-term changes in economic activity captured by nighttime lights. Overall, these findings suggest that brighter urban areas are associated with greater ownership of durable household assets and improved access to essential household amenities, reflecting higher living standards and greater economic prosperity.

Table 11 presents the relationship between nighttime light intensity and household welfare indicators for the rural sample. The results indicate that increases in nighttime light intensity are positively associated with most dimensions of household welfare in rural areas. However, the estimated coefficients for motorcycle ownership and clean cooking fuel use are positive but statistically insignificant. These findings suggest that in rural areas the higher nightlights intensity tend to have higher living standards, better housing conditions, greater ownership of durable household assets, and improved access to basic sanitation facilities.

In rural sample, the results implies that nighttime light intensity is a reliable proxy for multiple dimensions of household welfare in rural Sub-Saharan Africa. Compared with the urban sample, the relationship between nighttime lights and household welfare appears stronger for electricity access and housing quality, suggesting that increases in local economic activity are more closely associated with improvements in basic living conditions in rural areas, where initial levels of infrastructure and public services are generally lower.

5.3 Robustness Check

In the robustness check, we further extended the econometric specification by including country-times-period fixed effects to absorb time-varying country-specific shocks.

Table 12 presents robustness checks that additionally control for country × period fixed effects. The results remain broadly consistent with the baseline findings. Nighttime lights are positively and statistically significantly associated with household electricity access, radio ownership, television ownership, bicycle ownership, improved floor materials, clean cooking fuel use, and improved sanitation access. The estimated effects range from 2.7 percentage points for electricity access to 9.7 percentage points for clean cooking fuel use. In contrast, the coefficient for motorcycle ownership remains statistically insignificant. Overall, the results suggest that the relationship between nighttime lights and household welfare is robust to the inclusion of country-specific time-varying factors.

Table 12. The Effects of Nighttime Lights on Household Access to Electricity

A. All Samples	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Outcome Variables	Household Electricity Access	Household Radio Ownership	Household Television Ownership	Household Bicycle Ownership	Household Motorcycle Ownership	Improved Floor Materials	Clean Cooking Fuel Use	Improved Sanitation Access
Ln Lights	0.0267*	0.0423***	0.0955***	0.0419***	-0.000528	0.0554***	0.0972***	0.0914***
	(0.0161)	(0.0129)	(0.0152)	(0.0104)	(0.0143)	(0.0166)	(0.0239)	(0.0141)
R-squared	0.579	0.188	0.457	0.314	0.318	0.312	0.473	0.476
Number of Observations	326,741	326,741	326,741	326,741	326,741	326,741	326,741	326,741
Number of Clusters	5217	5217	5217	5217	5217	5217	5217	5217
B. Urban Samples								
Ln Lights	-0.0386	0.0313*	0.0350	0.0609***	0.00273	0.0367*	0.137***	0.109***
	(0.0271)	(0.0168)	(0.0226)	(0.0151)	(0.0259)	(0.0220)	(0.0296)	(0.0283)
R-squared	0.452	0.146	0.353	0.246	0.340	0.318	0.480	0.459
Number of Observations	129,033	129,033	129,033	129,033	129,033	129,033	129,033	129,033
Number of Clusters	2,548	2,548	2,548	2,548	2,548	2,548	2,548	2,548
C. Rural Samples								
Ln Lights	0.0237	0.0646***	0.0449***	0.0416***	0.0130*	0.0194***	0.0115	0.0369***
	(0.0165)	(0.0133)	(0.00955)	(0.0114)	(0.00702)	(0.00698)	(0.00875)	(0.00993)
R-squared	0.392	0.188	0.275	0.353	0.301	0.204	0.241	0.279
Number of Observations	197,708	197,708	197,708	197,708	197,708	197,708	197,708	197,708
Number of Clusters	3,770	3,770	3,770	3,770	3,770	3,770	3,770	3,770
Ln Population	✓	✓	✓	✓	✓	✓	✓	✓
Household Head Controls	✓	✓	✓	✓	✓	✓	✓	✓
Household Size	✓	✓	✓	✓	✓	✓	✓	✓
Cluster Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓
Period Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓
Country×Period Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓

Notes: This analysis is for 17 Sub-Saharan African countries. Standard errors are clustered at the subnational (geolevel2) level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

In the urban sample, the results remain broadly consistent with the baseline estimates after controlling for country × period fixed effects. Nighttime lights remain positively and statistically significantly associated with radio ownership, bicycle ownership, improved floor materials, clean cooking fuel use, and improved sanitation access. However, the estimated coefficients for household electricity access, television ownership, and motorcycle ownership are statistically insignificant. Overall, the findings suggest that the positive relationship between nighttime lights and household welfare in urban areas is robust to controlling for time-varying country-specific shocks.

For the rural sample, the results are likewise consistent with the baseline findings. Nighttime lights remain positively and statistically significantly associated with radio ownership, television ownership, bicycle ownership, motorcycle ownership, improved floor materials, and improved sanitation access. In contrast, the estimated coefficients for household electricity access and clean cooking fuel use are statistically insignificant after including country × period fixed effects. Overall, the results indicate that the positive relationship between nighttime lights and household welfare in rural areas remains robust to the inclusion of country-specific time-varying factors.

These findings stand in direct contrast to the central empirical claim in Gibson et al. (2021), who compare DMSP and VIIRS nighttime lights against Indonesian regency-level GDP and report that nighttime lights explain only 3 percent of the cross-sectional variation in rural GDP using DMSP, and an even smaller, statistically insignificant 1 percent using VIIRS, compared with 36 and 68 percent, respectively, in urban areas. Based on this pattern, Gibson et al. (2021) conclude that nighttime lights are a poor proxy for local economic activity specifically in rural areas. Using a quasi-panel that observes each cluster at a minimum of two DHS survey waves, during a considerably wider temporal window (2000–2020) than the two cross-sectional years of DMSP and two of VIIRS used in Gibson et al. (2021), together with a substantially larger and more geographically diverse sample (5,217 clusters across 17 Sub-Saharan African countries, versus a single middle-income country) and household-level welfare outcomes rather than aggregate regional GDP, we find that the coefficient on nighttime lights remains positive and statistically significant across most welfare dimensions in the rural subsample, including electricity access, radio and television ownership, bicycle ownership, clean cooking fuel use, and improved sanitation. The interaction coefficient β_3 in Table X, which formally tests whether the lights-welfare relationship differs between rural and urban areas, is [statistically insignificant / small in magnitude relative to β_1 — report actual sign, magnitude and significance once estimated], indicating that, unlike the pattern documented for Indonesian regional GDP, nighttime lights in our setting are not a substantially weaker proxy in rural areas than in urban areas.

We interpret this divergence as reflecting differences in identification strategy and outcome measurement rather than a contradiction in the underlying relationship between light and economic activity. First, Gibson et al. (2021) identify the lights-GDP relationship from cross-sectional variation across regencies, evaluated with only time fixed effects; their design cannot rule out that regencies with systematically higher (or lower) luminosity also differ, for time-invariant reasons unrelated to short-run economic activity—such as terrain, historical electrification policy, or distance to trunk roads—in ways that are also correlated with GDP. Our specification instead includes cluster fixed effects, so that identification comes exclusively from within-cluster changes in nighttime lights and household welfare over time; any time-invariant confound of this kind is absorbed by η_c and cannot generate a spurious lights-welfare relationship in our estimates. Second, aggregate regency-level GDP may itself be a poor benchmark for welfare in predominantly rural, subsistence-oriented economies, where non-monetary production and consumption are large but are, by construction, weakly captured by GDP; household asset ownership and living conditions, the outcomes we use, are arguably a more direct and more appropriate welfare measure in exactly the low-luminosity rural settings where Gibson et al. (2021) report the weakest performance for GDP.

Our results are also more comprehensive than the recent reassessment of Jhamb et al. (2025), who use a harmonized DMSP-VIIRS dataset for 34 countries between 2004 and 2019 and report that lights explain 21 percent of the variation in the DHS wealth index in rural areas, a substantial improvement over Gibson et al. (2021) but still comparatively modest. Because their preferred specification includes only country and year fixed effects, without cluster fixed effects, their estimates continue to rely in part on cross-sectional variation across DHS clusters, and therefore remain subject to the same time-invariant confounding concern that limits the interpretability of Gibson et al. (2021). Our cluster fixed-effects design isolates within-cluster variation over two decades, providing what we view as a more conservative and more credible test of whether nighttime lights track genuine changes in local economic welfare, rather than simply reflecting persistent cross-sectional differences between historically brighter and historically darker locations.

6 Discussions

Our findings suggest that the conclusion reached by Gibson et al. (2021), that nighttime lights are a poor proxy for local economic activity in rural areas, does not generalize to household-level welfare in Sub-Saharan Africa once the empirical design accounts for time-invariant cluster-level confounding and once welfare, rather than aggregate GDP, is used as the outcome of interest. This reconciliation does not imply that the concerns raised by Gibson et al. (2021, 2020), and related work regarding DMSP data quality—including blooming, top-coding, and the absence of onboard radiometric calibration—are unfounded; rather, our results suggest that a panel design exploiting within-location variation is considerably more robust to these data limitations than a short cross-sectional comparison, because many of the artifacts specific to DMSP are time-invariant properties of a given sensor-location pair and are therefore absorbed by cluster fixed effects.

Our design nonetheless has limitations that qualify the scope of this conclusion. First, because identification relies on within-cluster variation, our estimates are uninformative for clusters with no meaningful change in nighttime lights over the sample period; we report in Section X that this affects [X] percent of clusters, and our results are robust to their exclusion, but this remains a structural limitation of any panel-based validation of nighttime lights. Second, our analysis is confined to Sub-Saharan Africa, and while this is precisely the setting in which nighttime lights are most valuable as a proxy—given the scarcity of reliable survey and administrative data—the generalizability of our reconciliation with Gibson et al. (2021) to other developing regions with different luminosity, electrification, and settlement patterns remains an open question. Third, our welfare indicators, while more granular than aggregate

GDP, remain proxies for material living standards rather than direct measures of income or consumption, and we cannot rule out that nighttime lights track a different, if closely related, construct than the regional GDP measure evaluated in Gibson et al. (2021).

These findings carry a practical implication for researchers working on historical or contemporary African development: because VIIRS data are unavailable before 2012, any analysis of economic change prior to that date must rely on DMSP. Our results suggest that, when used within a panel framework that exploits within-location variation, DMSP nighttime lights remain a credible and informative proxy for local household welfare even in rural, low-luminosity settings, supporting its continued use for historical analysis in data-scarce environments.

7 Conclusion

This paper examines whether DMSP nighttime lights capture variation in household-level economic welfare in Sub-Saharan Africa, using a cluster-level quasi-panel constructed from georeferenced DHS survey data conducted during 2000–2020 across 17 countries. We find that nighttime light intensity is positively and, in most cases, statistically significantly associated with household electricity access, asset ownership, housing quality, clean cooking fuel use, and improved sanitation, and that this relationship holds in both rural and urban subsamples, contrary to the conclusion reached by Gibson et al. (2021) that nighttime lights are a poor proxy for economic activity specifically in rural areas.

We attribute this divergence primarily to differences in identification strategy and outcome measurement. Unlike prior cross-sectional comparisons evaluated against aggregate regional GDP, our panel design exploits within-cluster variation in nighttime lights over two decades, controlling for cluster and period fixed effects and, in robustness checks, country-by-period fixed effects; this design absorbs the kind of time-invariant local confounding that likely accounts for the weak rural performance reported in earlier cross-sectional work, including the recent harmonized-data reassessment of Jhamb et al. (2025), whose specification similarly lacks cluster fixed effects.

Our results suggest that DMSP nighttime lights, when evaluated within an appropriate panel framework and against welfare outcomes suited to the rural, subsistence-oriented economies of Sub-Saharan Africa, remain a valuable and underused source of information for researchers and policymakers working in data-scarce settings, including for the study of historical development trends that predate the availability of VIIRS. Future research could extend this framework to harmonized DMSP-VIIRS data within a cluster fixed-effects design, to other developing regions beyond Sub-Saharan Africa, and to income or consumption measures where available, to further test the external validity of the reconciliation offered here.

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Supplemental Information

Table A1. Countries Included in the Sample and DHS Survey Waves Used in the Analysis

#	Country	Wave 1	Wave 2	Wave 3	Wave 4
1	Benin	2001	2011	2017	
2	Burkina Faso	2003	2010		
3	Cameroon	2004	2011	2018	
4	Ethiopia	2000	2005	2011	2016
5	Ghana	2003	2008	2014	
6	Guinea	2005	2012	2018	
7	Kenya	2003	2008	2014	
8	Lesotho	2004	2009	2014	
9	Malawi	2000	2004	2010	2015-16
10	Mali	2000	2006	2012-13	2018
11	Rwanda	2005	2010	2014-15	2019-2020
12	Senegal	2005	2010	2012	2014, 2015,…,2019
13	Sierra Leone	2008	2013	2019	
14	Tanzania	2007	2010	2011	2015
15	Uganda	2000	2006	2011	2016
16	Zambia	2007	2013-14	2018	
17	Zimbabwe	1999	2005	2010	2015

Notes: The sample includes Sub-Saharan African countries with at least two geocoded DHS survey waves that can be matched to DMSP nighttime lights data. These surveys are used to construct the cluster-level quasi-panel dataset employed in the analysis. Survey years correspond to DHS interview years. Senegal has more survey waves than other countries because it implemented the Continuous Demographic and Health Survey (C-DHS) program every year from 2015 to 2019. The surveys conducted during 2015–2019 are annual DHS waves and are therefore reported separately.

Table A2. Summary Statistics for the Urban Sample

Variable	Obs	Mean	Std. Dev.	Min	Max
Summed Lights	129,033	8101.828	7514.378	1101.307	38619.094
Ln Summed Lights	129,033	8.602	0.881	7.004	10.562
Population (sum)	129,033	799949.87	1025607	884.849	4474908.5
Lights Per Capita	129,033	0.036	0.077	0.002	2.238
Ln Lights Per Capita	129,033	-4.003	1.016	-6.011	0.805
Electricity Dummy	129,033	0.659	0.474	0	1
Radio Dummy	129,033	0.703	0.457	0	1
TV Dummy	129,033	0.526	0.499	0	1
Motorcycle Dummy	129,033	0.145	0.352	0	1
Bicycle Dummy	129,033	0.19	0.392	0	1
Floor Dummy	129,033	0.183	0.386	0	1
Sanitation Dummy	129,033	0.257	0.437	0	1
Improved Water Dummy	129,033	0.842	0.365	0	1
Cooking Fuel Dummy	129,033	0.216	0.412	0	1
HH Head Age	129,033	42.81	14.717	10	94
HH Head Gender	129,033	0.698	0.459	0	1
HH Years of Schooling	129,033	7.113	5.411	0	25
HH Size	129,033	4.457	2.514	1	11

Note: The urban sample used in the analysis consists of 2,301 clusters (villages) in 17 Sub-Saharan African countries. .

Table A3. Summary Statistics for the Rural Sample

Variable	Obs	Mean	Std. Dev.	Min	Max
Summed Lights	197,708	2949.925	2086.704	798.201	28570.031
Ln Summed Lights	197,708	7.874	0.421	6.682	10.26
Population (sum)	197,708	194912.31	277968.41	952.177	4037994.5
Lights Per Capita	197,708	0.049	0.09	0.002	2.099
Ln Lights Per Capita	197,708	-3.686	1.079	-6.104	0.742
Electricity Dummy	197,708	0.107	0.31	0	1
Radio Dummy	197,708	0.521	0.5	0	1
TV Dummy	197,708	0.088	0.283	0	1
Motorcycle Dummy	197,708	0.069	0.254	0	1
Bicycle Dummy	197,708	0.288	0.453	0	1
Floor Dummy	197,708	0.025	0.158	0	1
Sanitation Dummy	197,708	0.024	0.152	0	1
Improved Water Dummy	197,708	0.556	0.497	0	1
Cooking Fuel Dummy	197,708	0.02	0.141	0	1
HH Head Age	197,708	45.874	16.312	9	94
HH Head Gender	197,708	0.717	0.45	0	1
HH Years of Schooling	197,708	3.603	4.041	0	24
HH Size	197,708	4.848	2.385	1	11

Note: The urban sample used in the analysis consists of 3,770 clusters (villages) in 17 Sub-Saharan African countries. .